Theory of Sliding ^{4}He Atoms on Graphite Substrate

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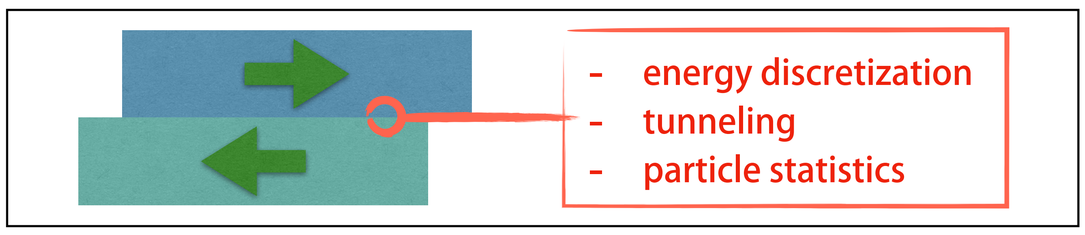
Introduction

QCM Tribology

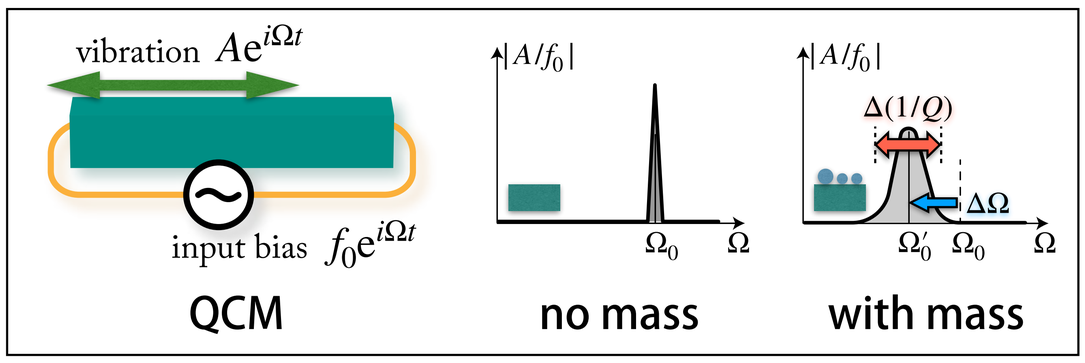
Understanding the nature of sliding friction from the **microscopic** and **quantum** viewpoints is required to treat the friction in micro/nanomechanical systems (e.g. MEMS/NEMS).



Quartz Crystal Microbalance (**QCM**) is one of the most sophisticated experimental tools, which captures dynamic/static friction in molecularly thin layers of matter by measuring

- changes in the resonant frequency Ω
- changes in the quality factor (Q-value) ($1/Q$)

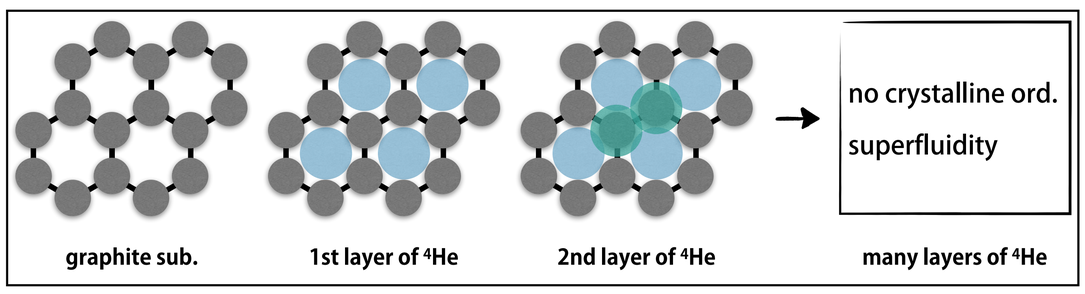
from the mass on the quartz.



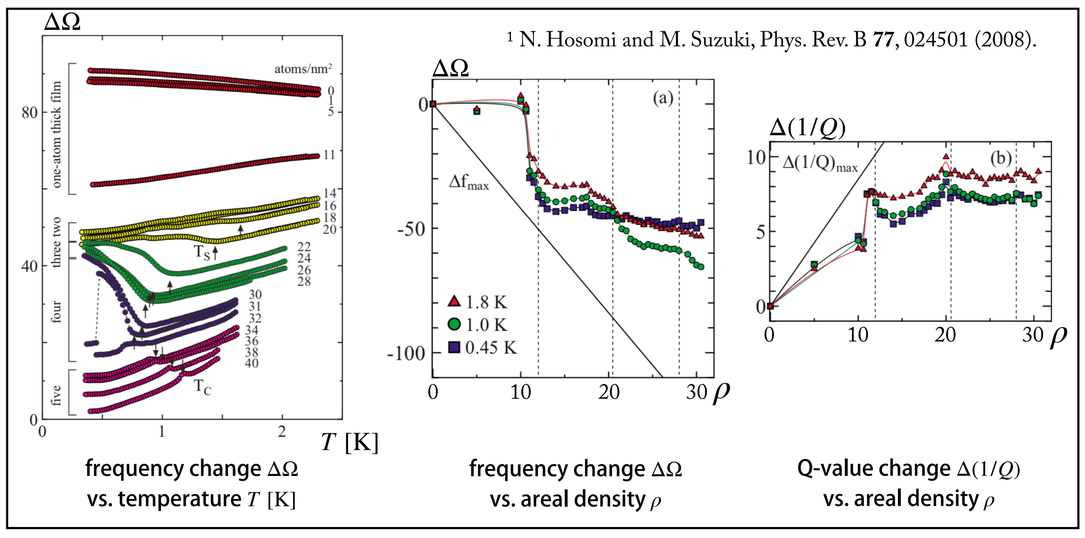
An example: ^{4}He on Graphite

Helium 4 (^{4}He) atoms adsorbed graphite substrate (or a single graphene sheet) are known to form so-called "**two-dimensional ^{4}He**" state exhibiting

- crystalline orders (when the system is almost quasi-two-dimensional),
- superfluidity like the bulk three-dimensional ^{4}He (when the particles increase).



The non-equilibrium properties of two-dimensional ^{4}He was investigated using QCM technique and vibration-induced decoupling of the ^{4}He atoms and the graphite substrate was reported, but properties of the many-layers case is almost unknown.



[1] N. Hosomi and M. Suzuki, Phys. Rev. B **77**, 024501 (2008).

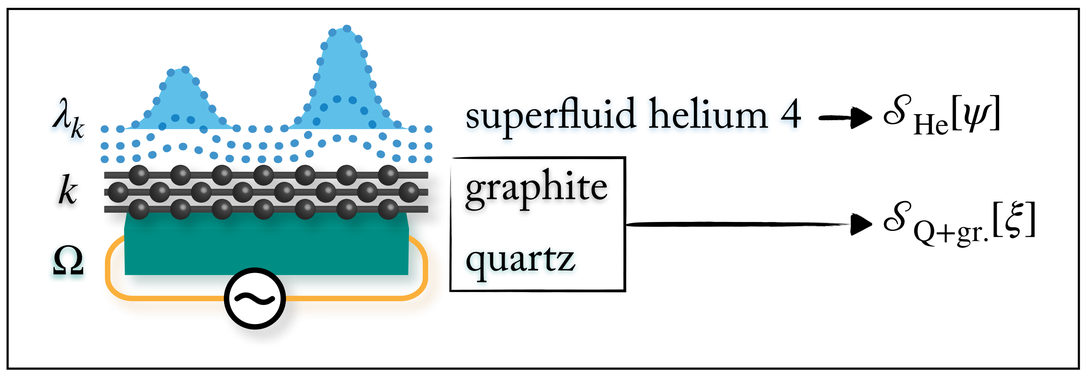
Fully-quantum description of ^{4}He atoms on graphite substrate might be essential.

Methods and Results

Coupled Model

We construct a coupled model of superfluid helium ^{4}He, graphite and QCM to investigate the quantum aspect of sliding ^{4}He on the graphite by introducing the action of each subsystem, namely $\mathcal{S}_{\text{Q+gr.}}[\xi]$ and $\mathcal{S}_{\text{He}}[\psi]$.

The quartz has its own **complex resonant frequency** Ω as in the case of the experiment and transmits its vibration into the upper graphite. Due to the intrinsic phonon band, the graphite converts the transmitted vibration into a spatial wave with a specific wave number k and excites the corresponding energy λ_k of ^{4}He. We regard the changes in the frequency and the Q-value in the experiment as the effect of **renormalization** due to the second process of the Ω-k-λ_k conversion. We use the equation for a set of harmonic oscillators for $\mathcal{S}_{\text{Q+gr.}}[\xi]$ and use the Gross-Pitaevskii equation and the Bogoliubov equation for $\mathcal{S}_{\text{He}}[\psi]$.



Effective Action Formalism

We describe the renormalized effect of the quartz and the graphite subsystems out of the whole system by an **effective action**. The effective action is defined by the equality for the partition function $\mathcal{Z}_{tot}$ of the whole system with a coupling action $\mathcal{S}_c[\xi,\psi]$.

the action of the system

$$\mathcal{S}_{\text{tot}}[\xi,\psi] := \mathcal{S}_{\text{Q+gr.}}[\xi] + \mathcal{S}_c[\xi,\psi] + \mathcal{S}_{\text{He}}[\psi]$$

linear both in ξ and ψ

partition function

$$\mathcal{Z}_{tot} := \int \mathcal{D}\xi\mathcal{D}\psi \exp\left[i\mathcal{S}_{\text{tot}}[\xi,\psi]\right] = \int \mathcal{D}\xi \exp\left[i\mathcal{S}_{\text{eff}}^{(\text{Q+gr.})}[\xi]\right]$$

the effective action for the graphite-quartz

Using the effective action $\mathcal{S}_{\text{eff}}^{(\text{Q+gr.})}[\xi]$, we derive the condition for the renormalized frequencies $\{\omega_k'\}$ by the variational principle.
The resultant equation for the renormalized complex frequency $\Omega = \omega_k'$ has an anomalous term which leads to a simple **algebraic equation**.

$$\Delta\Omega \sim \Re[\omega_k' - \omega_k]$$

$$\Delta(1/Q) \sim \Im[\omega_k' - \omega_k]$$

variation

$$\mathcal{S}_{\text{Q+gr.}}[\xi] \Longrightarrow \xi_k(t) = \xi_k(0)e^{-i\omega_k t}$$

renormalization

$$\mathcal{S}_{\text{eff}}^{(\text{Q+gr.})}[\xi] \Longrightarrow \xi_k(t) = \xi_k(0)e^{-i\omega_k' t}$$

variation

$$\delta\mathcal{S}_{\text{eff}}^{(\text{Q+gr.})}/\delta\xi = 0 \longrightarrow \Omega = \omega_k - \frac{I_k|f_k|^2}{2}\frac{\hbar\Omega(\Omega+\omega_k)}{\lambda_k - (\hbar\Omega)^2}$$

$$I_k := \frac{(\epsilon_k - gn_0)^2 + 2gn_0(\epsilon_k - gn_0) + 2(gn_0)^2}{(\epsilon_k - gn_0)^2 + 4gn_0(\epsilon_k - gn_0) + 3(gn_0)^2}$$

$$\lambda_k := \sqrt{\epsilon_k^2 + 2gn_0\epsilon_k}$$

Conclusion

We derived an equation for the renormalized complex frequency $\{\omega_k'\}$, which expresses both the resonance peak and its broadening as its real and imaginary parts. This analysis may reveal the quantum nature of ^{4}He atoms sliding on graphite substrate. For more detail analysis, we have to solve the equation with a set of actual physical parameters.